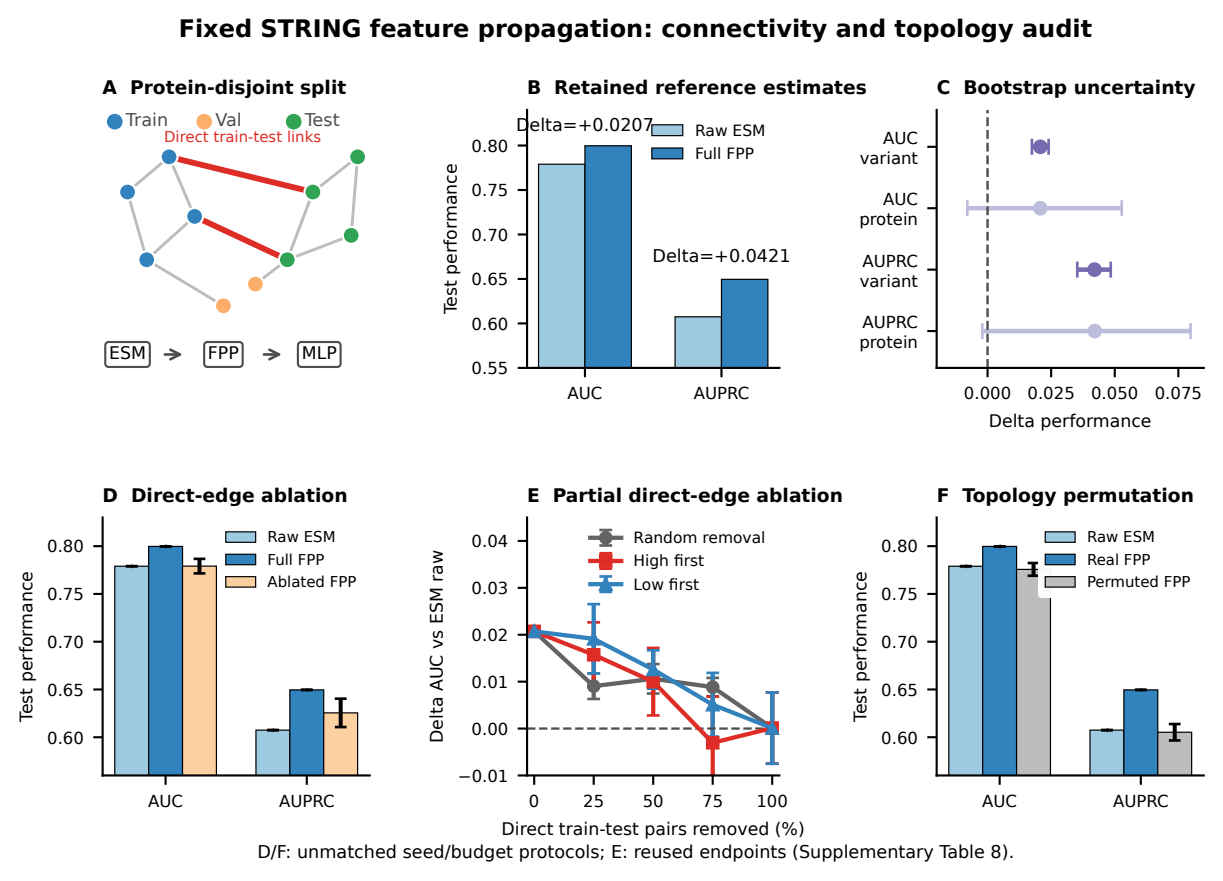
Fixed STRING feature propagation: connectivity and topology audit
A Protein-disjoint split
Train
Val
Test
Direct train-test links
ESM
FPP
MLP
B Retained reference estimates
Delta=+0.0207
Delta=+0.0421
Raw ESM
Full FPP
Test performance
0.80
0.75
0.70
0.65
0.60
0.55
AUC
AUPRC
C Bootstrap uncertainty
AUC variant
AUC protein
AUPRC variant
AUPRC protein
0.000
0.025
0.050
0.075
Delta performance
D Direct-edge ablation
Raw ESM
Full FPP
Ablated FPP
Test performance
0.80
0.75
0.70
0.65
0.60
AUC
AUPRC
E Partial direct-edge ablation
Random removal
High first
Low first
Delta AUC vs ESM raw
0.04
0.03
0.02
0.01
0.00
−0.01
0
25
50
75
100
Direct train-test pairs removed (%)
F Topology permutation
Raw ESM
Real FPP
Permuted FPP
Test performance
0.80
0.75
0.70
0.65
0.60
AUC
AUPRC
D/F: unmatched seed/budget protocols; E: reused endpoints (Supplementary Table 8).